JDBC Theory Questions:

1. What is RowSet?
2. Difference between JdbcRowSet and CachedRowSet?
3. What is connection pooling advantages?
4. What is fetch size?
5. What is pagination?
6. How to improve performance?
7. How to handle large data?
8. How to use stored procedures?
9. How to use batch updates?
10. How to reuse connections?

MCQ Questions

1.

```
Connection con = DriverManager.getConnection(url, user, pass);
System.out.println(con != null);
```

A. true
B. false
C. Error
D. null

2.

```
Statement st = con.createStatement();
int x = st.executeUpdate("INSERT INTO emp VALUES(1,'A')");
System.out.println(x);
```

A. 0
B. 1
C. true
D. Error

3.

```
ResultSet rs = st.executeQuery("SELECT * FROM emp");
while(rs.next()){
    System.out.print(rs.getInt(1));
}
```

A. Error
B. Prints first column values
C. Prints all columns
D. Nothing

4.

```
PreparedStatement ps = con.prepareStatement("INSERT INTO emp VALUES(?,?)");
ps.setInt(1,1);
ps.setString(2,"A");
ps.executeUpdate();
```

A. Error
B. Inserts record
C. Prints data
D. Nothing

5.

```
System.out.println(rs.next());
```

A. true/false
B. row data
C. error
D. null

6.

```
con.setAutoCommit(false);
```

A. Enables auto commit
B. Disables auto commit
C. Error
D. Commit happens automatically

7.

```
con.rollback();
```

A. Saves data
B. Undo changes
C. Deletes table
D. Error

8.

```
con.commit();
```

A. Undo
B. Save changes
C. Error
D. Delete

9.

```
ps.executeQuery();
```

A. Used for SELECT
B. Used for INSERT
C. Error always
D. Delete

10.

```
con.close();
```

A. Opens connection
B. Closes connection
C. Deletes DB
D. Error

Practice Programs

1) **Call a stored procedure from Java using JDBC.**

Task:

- Create a stored procedure in DB (e.g., get student by ID)
- Call it using `CallableStatement`

Output:
Display returned record

2) **Perform multiple SQL operations in a single batch.**

Task:

- Insert/update multiple records
- Execute using `executeBatch()`

Output:
All queries executed successfully

3) **Display records page-wise from database.**

Task:

- Fetch limited rows using LIMIT
- Show 5 records per page

Example:
Page 1 → records 1–5
Page 2 → records 6–10

4) **Efficiently retrieve large datasets from database.**

Task:

- Use `setFetchSize()`
- Avoid loading entire data into memory

5) **Reuse database connections instead of creating new ones.**

Task:

- Implement basic connection pool logic
- Reuse connection for multiple queries

6) **Improve performance of database operations.**

Task:

- Use batch processing
- Use PreparedStatement
- Optimize queries

7) **Transfer money between two accounts.**

Task:

- Deduct from sender
- Add to receiver
- Use commit/rollback

Condition:
If any error → rollback

8) **Display large data in pages.**

Task:

- Use LIMIT and OFFSET
- Show user-friendly navigation

9) **Fetch DB records and store them in CSV format.**
10) **Read CSV file and insert data into DB.**